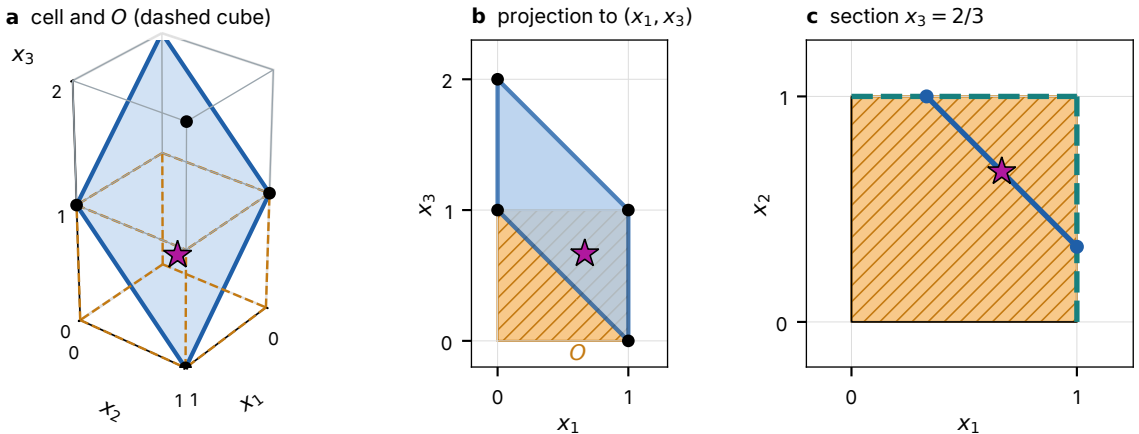
Scaled $E_3 \subset \mathbf{Z}^3$, box (1, 1, 2): the 2-cell $\Pi((0,0,2);(0,1,-1),(1,0,-1))$ meets O at $(\frac{2}{3},\frac{2}{3},\frac{2}{3})$



(c): blue = section of the cell; teal dashed = $U(\mathbf{Z}^3, Q)$ at height 2/3 (the lines $x_1 = 1$, $x_2 = 1$ only).
The section meets them only at its end points; its interior lies in O.